

Personalized LLM-Based Smart Home Assistant Using the TIAGo Robot: A ROS-Centric Framework for Human-Robot Interaction

Human-Robot Interaction Course Project

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Abstract—The integration of Large Language Models (LLMs) with robotics frameworks such as the Robot Operating System (ROS) represents a significant advancement in Human-Robot Interaction (HRI). This project focuses on developing a personalized home assistant robot using the TIAGo robotic platform, leveraging cutting-edge technologies like OpenAI Whisper for speech recognition, ChatGPT 4o model for reasoning and decision-making, and ElevenLabs for generating natural speech responses. The primary objective is to create a system that enables intuitive natural language interaction, adaptive task execution, and efficient collaboration with users in dynamic home environments. The proposed framework emphasizes modularity, scalability, and integration with state-of-the-art algorithms, enabling the TIAGo robot to perform tasks such as pick-and-place operations or personalized conversational interactions. By addressing challenges in user input understanding, context-aware actions, and environmental adaptability, this work aims to design and assess the usability and functionality of home assistant robots. Experimental results demonstrate the effectiveness of the proposed system in improving user experience and achieving efficient task execution in simulated environments.

Index Terms—Human-Robot Interaction, Large Language Models, TIAGo Robot, ROS, Speech Recognition, Context-Aware Robotics.

I. INTRODUCTION

THE integration of Large Language Models (LLMs) with robotic platforms marks a transformative step in the evolution of Human-Robot Interaction (HRI). As the demand for personalized, adaptive, and context-aware robotic systems increases, home assistant robots are gaining traction for their potential to simplify daily activities, enhance convenience, and improve the quality of life in modern households.

Recent advancements in NLP have demonstrated the potential of LLMs, like GPT-based architectures, to improve contextually relevant responses, enabling better communication and reasoning in dynamic environments. Studies such as Kennedy et al. [1] have highlighted the challenges of speech recognition, particularly for diverse user groups like children and non-native speakers, which emphasizes the need for adaptable systems. Robotic platforms like TIAGo have been widely used in tasks such as pick-and-place and human interaction, showcasing their flexibility with ROS and frameworks like MoveIt for manipulation [2]. The ROS-LLM project [3] has pioneered the integration of LLMs in the ROS ecosystem, enabling natural language-based interaction, and this work extends it by incorporating OpenAI Whisper [4] for advanced speech

recognition, fine-tuning the ChatGPT 4o model for reasoning, and using ElevenLabs for personalized speech output.

This project proposes a robust framework that combines these capabilities with the TIAGo robotic platform [5] to create a personalized home assistant robot. By leveraging OpenAI Whisper, ChatGPT 4o, and ElevenLabs [6], the system aims to enable seamless interactions between humans and robots, facilitating context-aware tasks and adaptability in dynamic home environments.

The significance of this work lies in addressing challenges such as accurate natural language understanding, adaptive task planning, and efficient execution of robotic actions. By integrating ROS as the underlying framework, the proposed system ensures modularity and scalability, enabling the TIAGo robot to perform a wide range of tasks, from conversational interactions to practical operations like pick-and-place.

The following sections outline the methodology, key components, and experimental results of the project, emphasizing its potential to advance the field of home assistant robotics.

II. EXPERIMENTAL DESIGN

This section outlines the experimental setup, including the robotic platform, the environment utilized for testing, and the types of human-robot interactions considered.

A. Simulation Environment

The experimental environment for this project is a simulated small house, designed to emulate realistic conditions found in home settings. This environment, provided by the AWS Robomaker Small House World, is used to test the robot's navigation, object manipulation, and human interaction abilities. The environment includes a kitchen, dining table, living room, and bedroom. The objects in the environment are a banana, books, apples, and a coffee mug.

B. Robotic Platform and Capabilities

The robotic platform employed in this study is the TIAGo mobile manipulator, which is designed for a wide range of tasks, including manipulation, navigation, and human-robot interaction. TIAGo is equipped with a 7-DOF arm, a mobile base, and multiple sensors, such as cameras and LiDAR, allowing it to perform various tasks autonomously. The robot is controlled using ROS (Robot Operating System), providing



Fig. 1: Simulation Environment

a modular framework for task execution and control. TIAGo's capabilities in manipulation and perception, combined with advanced motion planning, make it an ideal candidate for this research.



Fig. 2: TIAGo mobile manipulator

C. Human-Robot Interaction Framework

Effective Human-Robot Interaction (HRI) is a crucial component of this project, enabling intuitive and natural communication between users and the TIAGo robot. The following subsections detail the key HRI methodologies employed:

1) *Voice Command Interaction*: The TIAGo robot is designed to recognize and process human voice commands, allowing users to interact with it naturally. A speech recognition module is integrated into the system using OpenAI Whisper, enabling the robot to interpret user instructions in real time. These commands include navigation tasks (e.g., "go to the kitchen"), object manipulation requests (e.g., "pick up the bottle"), and general interaction phrases.

2) *Robot Voice Response*: To enhance user engagement and provide clear feedback, the robot responds using a text-to-speech (TTS) module. This ensures that users receive auditory confirmations for their commands, improving transparency and usability. The responses are generated dynamically based on task status, including success confirmations (e.g., "I have picked up the bottle") and error messages (e.g., "I couldn't find the requested object").

3) *Gesture-Based Feedback*: In response to human input, the robot provides non-verbal feedback through its arm movements. These gesture-based cues serve as an acknowledgment of received commands and indicate the robot's current state or intended actions. For instance, the robot may extend its arm to signal task initiation or wave to confirm a successfully completed instruction.

4) *Socially Aware Interaction*: To ensure smooth interaction in shared spaces, the robot adheres to socially aware navigation principles. It respects personal space by adjusting its trajectory to avoid abrupt movements near humans.

III. METHODOLOGY

This section outlines the methodology used in our system, which integrates an LLM-based robotic assistant with ROS. We built upon an existing implementation of ROS-LLM [3] and modified it for our application. The methodology is structured into three main components: LLM Input, LLM Node, LLM Output.

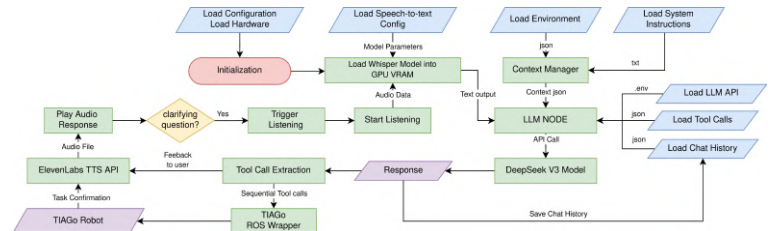


Fig. 3: Detailed Flowchart Design

A. LLM Input

The input to the LLM system is derived from user speech, which is processed using OpenAI Whisper. Whisper transcribes spoken language into text, serving as the input prompt to the LLM node. This transcription ensures accurate conversion of voice commands into actionable textual instructions that can be further processed.

B. LLM Node

The LLM Node is the core component that processes input text and determines appropriate robotic actions. This node is divided into several key components:

1) *Context*: The system maintains an understanding of the environment through a JSON file. This file contains:

- Object locations and their properties
- Room layouts and positions
- Mapping information for navigation

This environmental context enables the LLM to make informed decisions based on spatial awareness.

2) *System Prompt*: The system prompt defines the robot's identity, capabilities, and constraints. The robot, named Tiago, operates with the following specifications:

- Mobile manipulator with a single arm
- ROS-based movement and control
- Safe velocity limits (1.0 m/s linear, 1.0 rad/s angular)

- Interaction via natural conversation
- Execution of predefined motion sequences

The prompt also outlines interaction guidelines, including safety considerations, task execution constraints, and clarifying queries when instructions are ambiguous.

3) *Tool Calls*: The system employs a structured set of atomic actions and conversation responses.

a) *Robot Atomic Actions*: Atomic actions are fundamental operations that enable complex tasks. These include:

- **Publish command velocity** - Direct velocity commands for movement.
- **Move to point** - Navigate to a specific location.
- **Pick object** - Pick up objects.
- **Execute motion** - Perform a predefined movement pattern.
- **Speak to user** - Generate verbal feedback.
- **Place object** - Places objects at a designated position.

These atomic actions can be composed to execute complex behaviors, facilitating reusability and extensibility. The system allows for additional user-defined atomic actions through imitation learning.

b) *Conversation Response*: Apart from physical actions, the LLM can generate conversational outputs. These include:

- Providing answers to user queries
- Requesting clarification when the input is ambiguous
- Resuming active listening after incomplete instructions

4) *Chat History*: To ensure contextual continuity, the system maintains a record of previous interactions. This chat history allows the LLM to:

- Reference prior user commands
- Maintain task context across multiple exchanges
- Improve response relevance based on past inputs

C. LLM Output

The final stage involves executing the LLM's output through two primary modalities:

1) *Conversation Output*: To enable seamless interaction, the generated text response is converted into speech using ElevenLabs' text-to-speech synthesis. This ensures natural, vocalized feedback for users.

2) *LLM Robot Action Output*: For physical execution, the LLM translates its decisions into ROS-compatible commands. These commands may involve motion planning, object manipulation, or predefined gestures. The execution follows safe operation constraints and real-world feasibility checks.

IV. RESULTS AND DISCUSSION

The proposed system was evaluated within the simulated home environment (Figure 1) to assess its performance in executing atomic actions, long-horizon tasks, and user interactions, as well as the effectiveness of different large language models (LLMs) and speech recognition components.

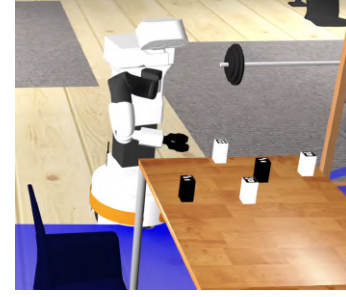


Fig. 4: Execution of the picking action by the TIAGo robot.

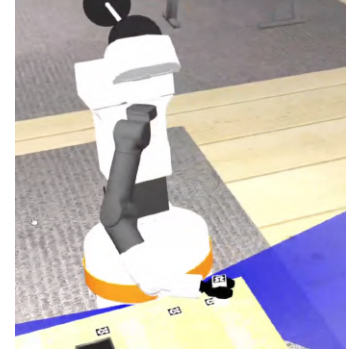


Fig. 5: Execution of the placing action by the TIAGo robot.

A. Evaluation of Atomic Actions

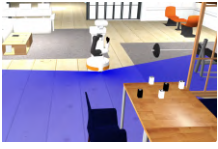
The system's ability to execute atomic actions was assessed by prompting the LLM to call specific functions corresponding to fundamental robotic operations. For instance, to evaluate the pick action, the LLM was prompted with the command "*pick up the banana*". The model correctly identified the appropriate function, **pick object**. Similarly, to test the navigation capability, the robot was prompted with "*go to the dining table*", successfully selecting the corresponding **move to point** action.

In all test cases, the robot accurately mapped natural language commands to the appropriate predefined actions. Figures 4 and 5 illustrate the execution of picking and placing actions within the environment.

B. Long-Horizon Task Execution

The ability of the system to perform long-horizon tasks, which require sequential execution of multiple atomic actions, was also evaluated. In one scenario, a book was placed in the environment, and the robot was prompted to "*fetch the book and bring it to the living room*". To successfully execute this task, the robot needed to navigate to the location of the book, pick it up, and transport it to the living room. The robot completed these steps autonomously without requiring additional context 7.

Similarly, another task involved placing a banana on the dining table and prompting the robot with "*Can you bring the banana and place it in the living room?*". Leveraging contextual knowledge about the banana's location, the robot autonomously navigated to the dining table, picked up the banana, transported it to the living room, and placed it appropriately (Figure 6). These results demonstrate the model's



(a) Step 1: Robot begins navigation towards the dining table.



(b) Step 2: Robot picks up the banana



(c) Step 3: Robot navigates to the living room with the banana.

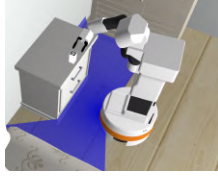


(d) Step 4: Robot places the banana in the living room.

Fig. 6: Sequence of actions taken by the robot to complete the task.



(a) Step 1: Robot navigates to the bedroom



(b) Step 2: Robot picks up the book



(c) Step 3: Robot moves to the specified location.

Fig. 7: Steps in the robot's action sequence.

capability to plan and execute complex tasks based solely on high-level user instructions.

C. Comparison of Different Large Language Models

The performance of different LLMs was compared using identical task prompts. Two models were evaluated: DeepSeek V3 from DeepSeek and OpenAI's ChatGPT-4o. The DeepSeek V3 model demonstrated superior performance in handling sequential and long-horizon tasks compared to ChatGPT-4o. For instance, when prompted to "bring the banana to the living room", ChatGPT-4o occasionally failed to execute the pick action after navigating to the banana's location.

Various prompt modifications, such as using "fetch" instead of "bring," did not significantly improve ChatGPT-4o's performance in this context. Conversely, DeepSeek V3 did not exhibit this issue, consistently executing the correct sequence of actions. This suggests that DeepSeek V3 has a more robust understanding of task planning in a home-assistant robotics context.

D. User Interaction and Feedback Mechanisms

To enhance human-robot interaction, the system incorporated user feedback mechanisms. The robot provided verbal confirmations before executing actions, during action execution, and upon task completion. Additionally, the ability to

request clarification from the user contributed to a more interactive and intuitive user experience. These features ensured that users remained informed about the robot's current state and intended actions, improving transparency and trust in the system.

E. Speech-to-Text Performance

Speech recognition was evaluated using OpenAI's Whisper model. The system exhibited high accuracy in transcribing spoken commands, with rare instances of misinterpretation. In cases where minor transcription errors occurred, the LLM successfully inferred the intended command based on contextual understanding. This highlights the advantage of integrating an LLM as a high-level planner, as it enhances the robustness of the system by mitigating minor errors in speech recognition.

V. CONCLUSION

This study presents a personalized LLM-based smart home assistant using the TIAGo robot, leveraging OpenAI Whisper for speech recognition, ChatGPT-4o for reasoning, and ElevenLabs for speech synthesis. The system successfully demonstrated its ability to interpret and execute both atomic and long-horizon tasks, ensuring a seamless interaction experience with users.

While certain simplifications were employed, such as the use of a predefined JSON file for specifying room and object locations, the system showcases significant potential in improving human-robot interactions in home environments. Experimental evaluations confirmed its effectiveness in executing complex commands autonomously and adapting to user interactions.

Future work will focus on enhancing task execution efficiency, improving semantic world understanding through deep learning models, and advancing personalization features to create more adaptive and intuitive home assistant robots. These improvements aim to further bridge the gap between human intent and robotic execution in real-world applications.

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